

**MATHEMATICS 2**

**40 MINUTES**

**INSTRUCTIONS TO CANDIDATES**

Please read these instructions carefully, but **do not open this question paper until you are ready to begin.**

This paper contains **27 multiple-choice questions**. There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all of the questions. Each question is worth one mark.

For each question, choose the one option you consider correct. If you make a mistake, erase thoroughly and try again.

You must complete the paper within the time limit.

You can use the question paper for rough working. Dictionaries and calculators are **NOT permitted**.

**Please wait until you are ready before turning this page.**

*Original ESAT CAMP practice material. Not an official UAT-UK or Pearson paper.*

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## **Mathematics 2**

- 1 The value of the definite integral

$$\int_1^2 \ln\left(\frac{e^{2x}}{x^2}\right) dx$$

is

- A 4
  - B  $2\ln 2$
  - C  $3 - 4\ln 2$
  - D  $5 - 4\ln 2$
  - E  $1 + 4\ln 2$
- 2 Find the number of solutions to the equation

$$\log_2(\sin x) + \log_2(3 \sin x) = 1 + \log_2(1 - \cos x)$$

for  $0 \leq x \leq 2\pi$ .

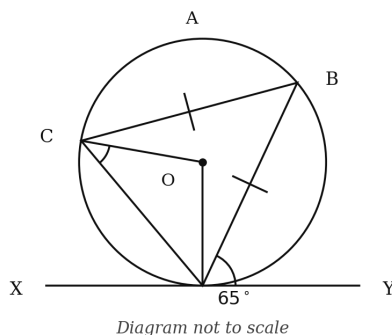
- A 0
- B 1
- C 2
- D 3
- E 4
- F 5

- 3** P, Q and R are regular polygons.  
 Q has twice as many sides as P.  
 R has three times as many sides as Q.  
 An interior angle of R is  $25^\circ$  larger than an interior angle of P.  
 How much larger is an interior angle of Q than an interior angle of P, in degrees?
- A** 5  
**B**  $8\frac{1}{3}$   
**C** 10  
**D**  $12\frac{1}{2}$   
**E** 15  
**F**  $16\frac{2}{3}$   
**G** 20
- 4** The function  $f(x) = 2x - x^2$  is defined for  $0 \leq x \leq k$ , where  $k > 2$ . Let  $I$  be the definite integral  $\int_0^k f(x) dx$ . Let  $A$  be the area of the region bounded by the curve  $y = f(x)$  and the  $x$ -axis for  $0 \leq x \leq k$ . Given that  $A = 3I$ , what is the value of  $k$ ?
- A** 2  
**B** 3  
**C**  $1 + \sqrt{2}$   
**D**  $1 + \sqrt{3}$   
**E**  $2 + \sqrt{3}$

- 5 Two vertices of a regular hexagon are at the points  $(1, 1)$  and  $(3, 5)$ .  
What is the sum of the areas of all possible regular hexagons with these points as vertices?

- A  $\frac{95\sqrt{3}}{2}$   
B  $\frac{75\sqrt{3}}{2}$   
C  $\frac{105\sqrt{3}}{2}$   
D  $40\sqrt{3}$   
E  $\frac{35\sqrt{3}}{2}$   
F  $\frac{95\sqrt{3}}{12}$

- 6 The straight line  $XY$  is tangent at point  $A$  to a circle with centre  $O$ .  
Points  $B$  and  $C$  lie on the circumference of the circle such that  $AB = BC$ .  
Angle  $YAB = 65^\circ$ .  
What is the size of angle  $OCA$ ?



- A  $25^\circ$   
B  $35^\circ$   
C  $40^\circ$   
D  $45^\circ$   
E  $50^\circ$   
F  $65^\circ$   
G  $75^\circ$   
H  $80^\circ$

- 7 The first term of an arithmetic sequence is 1 and its second term is  $\frac{1}{6}$ .

A geometric sequence has its first term equal to the third term of the arithmetic sequence, and its common ratio equal to the sum of the first three terms of the arithmetic sequence.

Find the sum to infinity of the geometric sequence.

- A  $-\frac{4}{3}$
- B 2
- C  $\frac{1}{3}$
- D  $-\frac{4}{9}$
- E  $-\frac{5}{3}$
- F  $-\frac{3}{4}$

- 8 The constant  $k$  is defined by

$$k = \int_0^2 (3x^2 + 1) dx.$$

A geometric series has first term  $k$  and common ratio  $1/3$ .

What is its sum to infinity?

- A 10
- B 12
- C 15
- D 18
- E 20
- F 30

- 9 What is the maximum value of the expression  $\sin^2 x + 3 \cos x$  for real values of  $x$ ?

- A 1
- B 3
- C  $13/4$
- D 4
- E 5

- 10** A particle moves along a straight line such that its displacement  $x$  in metres from a fixed origin at time  $t$  in seconds is given by:

$$x = 5 \cos(3t)$$

What is the total distance traveled by the particle between  $t = 0$  and  $t = \frac{5\pi}{6}$ ?

- A** 5
  - B** 15
  - C** 20
  - D** 25
  - E** 30
- 11** How many real solutions does the equation

$$\sin(2x) = \tan(x)$$

have in the interval  $0 \leq x \leq 2\pi$ ?

- A** 3
- B** 4
- C** 5
- D** 6
- E** 7



- 12 Two independent discrete random variables,  $X$  and  $Y$ , each take values in the set  $\{1, 2, 3\}$ .

The probability distributions of  $X$  and  $Y$  are given in the table below:

| $k$      | 1            | 2            | 3            |
|----------|--------------|--------------|--------------|
| $P(X=k)$ | <del>2</del> | <del>3</del> | <del>4</del> |
| $P(Y=k)$ | $p$          | $p$          | $q$          |

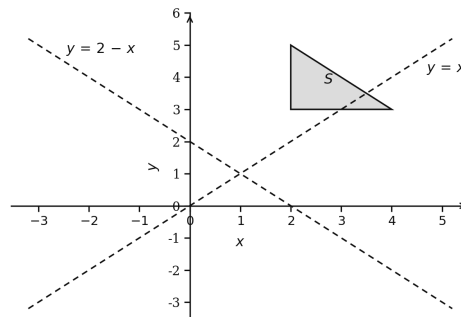
where  $p$  and  $q$  are constants.

Given that  $P(X + Y = 6) = \frac{1}{12}$ , what is the value of  $P(X + Y = 3)$ ?

- A  $\frac{1}{12}$
- B  $\frac{1}{8}$
- C  $\frac{1}{6}$
- D  $\frac{5}{24}$
- E  $\frac{1}{4}$
- F  $\frac{5}{12}$

- 13** The shape  $S$  shown in the diagram is reflected in the line  $y = x$ , and the resulting image is then reflected in the line  $y = 2 - x$ .

Which single transformation is equivalent to this sequence of two reflections?



- A** A translation by the vector  $\begin{pmatrix} 2 \\ 2 \end{pmatrix}$
- B** A translation by the vector  $\begin{pmatrix} -2 \\ -2 \end{pmatrix}$
- C** A reflection in the line  $x = 1$
- D** A reflection in the line  $y = 1$
- E** A rotation of  $90^\circ$  clockwise about the point  $(1, 1)$
- F** A rotation of  $90^\circ$  anticlockwise about the point  $(1, 1)$
- G** A rotation of  $180^\circ$  about the origin  $(0, 0)$
- H** A rotation of  $180^\circ$  about the point  $(1, 1)$
- 14** The area of the region enclosed by the curve  $y = 3x^2 + 1$ , the  $x$ -axis, and the lines  $x = 0$  and  $x = 2$ , is  $A$ .  
The area of the region enclosed by the line  $y = mx$ , the  $x$ -axis, and the lines  $x = 0$  and  $x = 2$ , is 10% of  $A$ .

What is the value of  $m$ ?

- A**  $\frac{1}{4}$
- B**  $\frac{2}{5}$
- C**  $\frac{1}{2}$
- D** 1
- E**  $\frac{13}{10}$
- F**  $\frac{9}{2}$

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What is the term independent of  $x$  in the expansion of  $(2x^2 - \frac{1}{x})^6$  ?

- A -160
- B -60
- C 15
- D 30
- E 60
- F 240

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The table shows summary statistics for the scores achieved by students in three different classes, X, Y, and Z.

| Class | Number of students | Mean score | Range of scores |
|-------|--------------------|------------|-----------------|
| X     | 15                 | 18         | 8               |
| Y     | 25                 | 24         | 14              |
| Z     | 10                 | 30         | 11              |

The results for all three classes are combined into a single dataset.

What can be deduced about the mean and range of the combined dataset?

- A mean = 23.4, range  $\leq 11$
- B mean = 23.4,  $11 < \text{range} < 14$
- C mean = 23.4, range  $\geq 14$
- D mean = 23.4, range  $\geq 33$
- E mean = 24.0, range  $\leq 11$
- F mean = 24.0,  $11 < \text{range} < 14$
- G mean = 24.0, range  $\geq 14$
- H mean = 24.0, range  $\geq 33$

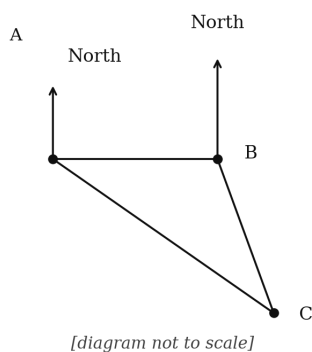
- 17 Three buoys,  $A$ ,  $B$ , and  $C$ , are floating in the sea as shown in the diagram.

Buoy  $B$  is due East of buoy  $A$ .

The distance from  $A$  to  $B$  is equal to the distance from  $B$  to  $C$ .

Buoy  $C$  is on a bearing of  $160^\circ$  from buoy  $B$ .

What is the bearing of buoy  $A$  from buoy  $C$ ?



- A  $015^\circ$
- B  $125^\circ$
- C  $285^\circ$
- D  $305^\circ$
- E  $325^\circ$
- F  $340^\circ$

- 18 A dataset of 25 integer values is summarised in the frequency table below:

| Value | Frequency |
|-------|-----------|
| 2     | $p$       |
| 4     | $q$       |
| 6     | 7         |
| 8     | $r$       |
| 10    | 2         |

where  $p$ ,  $q$ , and  $r$  are positive integers.

The dataset has:

- a mean of 5.44
- a median of 6
- a unique mode of 4

What is the value of  $p$ ?

- A 1
- B 2
- C 3
- D 4
- E 5
- F 7

- 19 A real number  $x$  satisfies the equation

$$\frac{e^{2x} - e^{-2x}}{e^{2x} + e^{-2x}} = \frac{1}{3}$$

What is the value of  $x$ ?

- A  $\frac{1}{4} \ln 2$
- B  $\frac{1}{2} \ln 2$
- C  $-\frac{1}{4} \ln 2$
- D  $-\frac{1}{2} \ln 2$
- E 0
- F  $\ln 2$

- 20** A container is constructed in the shape of a cuboid with a square base of side length  $x$  and height  $h$ . The container is open at the top. The total surface area of the container is fixed at a constant value  $S$ . Given that the volume of the container is maximised, which of the following equations correctly relates  $h$  and  $x$ ?

- A**  $h = \frac{1}{4}x$
- B**  $h = \frac{1}{2}x$
- C**  $h = x$
- D**  $h = 2x$
- E**  $h = 4x$

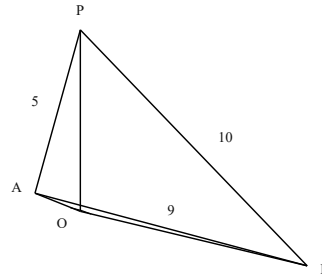
- 21** Find the coefficient of  $x^2$  in the expansion of

$$\left( \frac{x^3 - x}{x^2 - 1} + \frac{-2x^3 - 4x^2 - 2x - 4}{x^3 + 2x^2 + x + 2} \right)^5$$

- A**  $-80$
- B**  $-40$
- C**  $-20$
- D**  $20$
- E**  $40$
- F**  $80$

- 22** A vertical pole stands at a point O on horizontal ground. Points A and B are on the ground such that the triangle OAB is right-angled at O.

The distance from the top of the pole to A is 5, and the distance from the top of the pole to B is 10. The distance between A and B is 9.



What is the height of the pole?

- A**  $\sqrt{3}$
  - B**  $\sqrt{21}$
  - C**  $\sqrt{22}$
  - D**  $\sqrt{23}$
  - E**  $\sqrt{44}$
  - F**  $\sqrt{103}$
- 23** Three distinct numbers form the first three terms of an arithmetic progression and have a sum of 12. These three numbers, when reordered, form consecutive terms of a geometric progression. What is the value of the largest of the three numbers?
- A** 4
  - B** 8
  - C** 12
  - D** 16
  - E** 20

- 24** A curve has equation

$$y = \frac{k(2x^3)^2 - 4x^5}{x}$$

for  $x > 0$ , where  $k$  is a constant.

The gradient of the tangent to the curve at the point where  $x = 1$  is 12.

What is the value of  $k$ ?

- A**  $\frac{7}{5}$
- B**  $\frac{8}{5}$
- C**  $\frac{7}{4}$
- D**  $\frac{14}{5}$
- E** 4
- F** 8

- 25** What is the value of the sum

$$\sum_{n=2}^{100} \log_{10} \left( \frac{n^3 - n}{n^3 + n^2} \right)$$

- A** -2
- B** -1
- C** 0
- D** 1
- E** 2



- 26** The functions  $f(x)$  and  $g(x)$  satisfy the equation

$$2 \int_0^3 (f(x) + 1) \, dx + \int_0^3 (4g(x) - f(x)) \, dx = 15$$

What is the value of the following integral?

$$\int_0^3 (5f(x) + 20g(x) - 2x) \, dx$$

- A** 36
- B** 39
- C** 45
- D** 51
- E** 54
- F** 56

- 27** The solution set of the inequality

$$\frac{\log_2 x - k}{\log_2 x - 1} > 3$$

is the interval  $\frac{1}{4} < x < 2$ .

Find the value of the constant  $k$ .

- A** -1
- B** 1
- C** 3
- D** 4
- E** 5
- F** 7